

NEURON-LIKE Electronics



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Electronics based human body and brain activities are miraculous and a great feat by the creator. It provides a lesson for the pioneers of science and technology to learn and mimic these activities by artificially creating electronically-controlled devices.

The human brain is possibly the most complex entity in the Universe. It is absolutely remarkable and beautiful to contemplate, and the things you are capable of doing because of your brain are outstanding. The human brain is littered with some 100 billion nerve cells, together forming connections in tandem as each neuron is simultaneously engaged with another 1000 or so.

In total, some 20 million billion calculations per second are performed by the brain. Inspired by the operation and structure of the brain, engineers and scientists are now developing bio-inspired integrated circuit technology that mimics the neuron structure and operation of the human brain.

Although processors have gotten smaller and faster over time, only a few computers can compete with the speed and computing power of the human brain and none comes close to the organ's energy efficiency. So some engineers want to develop electronics that mimic how the brain computes to build more powerful and efficient devices.

Scientists have developed a new prototype for electronic synapses to replicate the human brain, which could one day make neural networks incredibly clever. At present, neural networks built by research groups around the world consist of computers that are being trained using complex computer algorithms to solve complex problems and gain a deeper recognition and understanding of art and the world around us.

Memristors

Memristors are the fourth class of electronic circuitry, alongside resistors, capacitors and inductors. Only confirmed to exist in 2008, memristors behave in a way similar to the synapses of neurons within the human brain. The resistance to current within a memristor is a product of currents that have previously flowed through it, meaning that the current flows easier as more current flows through.

Due to these properties, memristors hold potential for non-volatile memories, and can make computers better at understanding speech, images and the world around these.

In the past, it was very difficult to reproduce synapses because there are billions of neurons and thousands of synapses in the human brain, and the only way to get even remotely close to the power of a human brain using ordinary electronics is to utilise a gigantic amount of circuitry that would consume huge amounts of power. But then researchers managed to bring to life the concept of a memristor (using titanium oxide with two platinum electrodes), which means that conductivity in the memristor device changes depending on the charge passing through it, and connectivity can be changed from high to low.

This is similar to the way brain synapses work, and could make it possible for much more energy-efficient computer systems to be developed that have memories that retain information even when the power is off. So in the future, computers would not need to boot up and could be switched off and on like an electric light.

So now, researchers all over the world are trying to use memristors